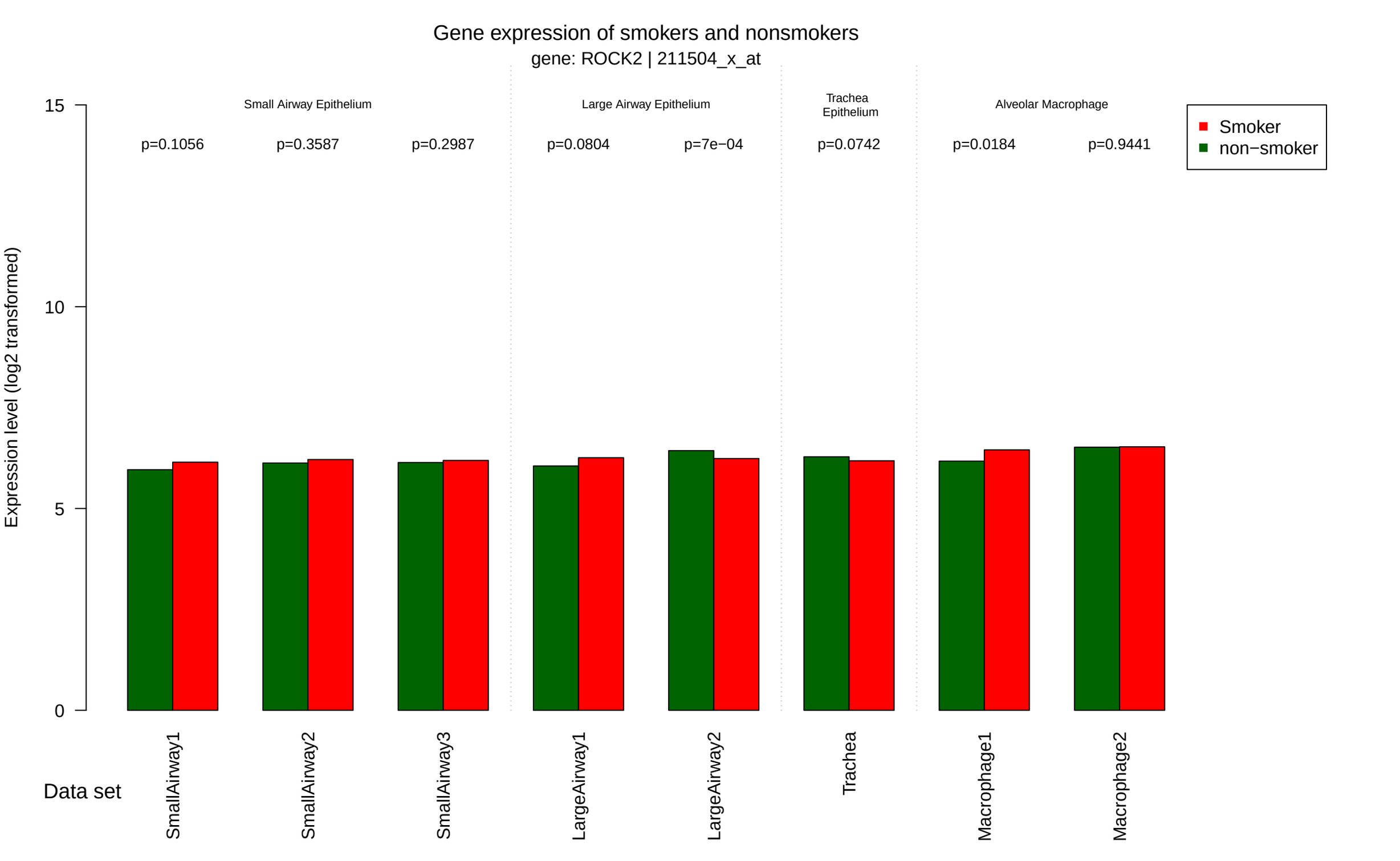
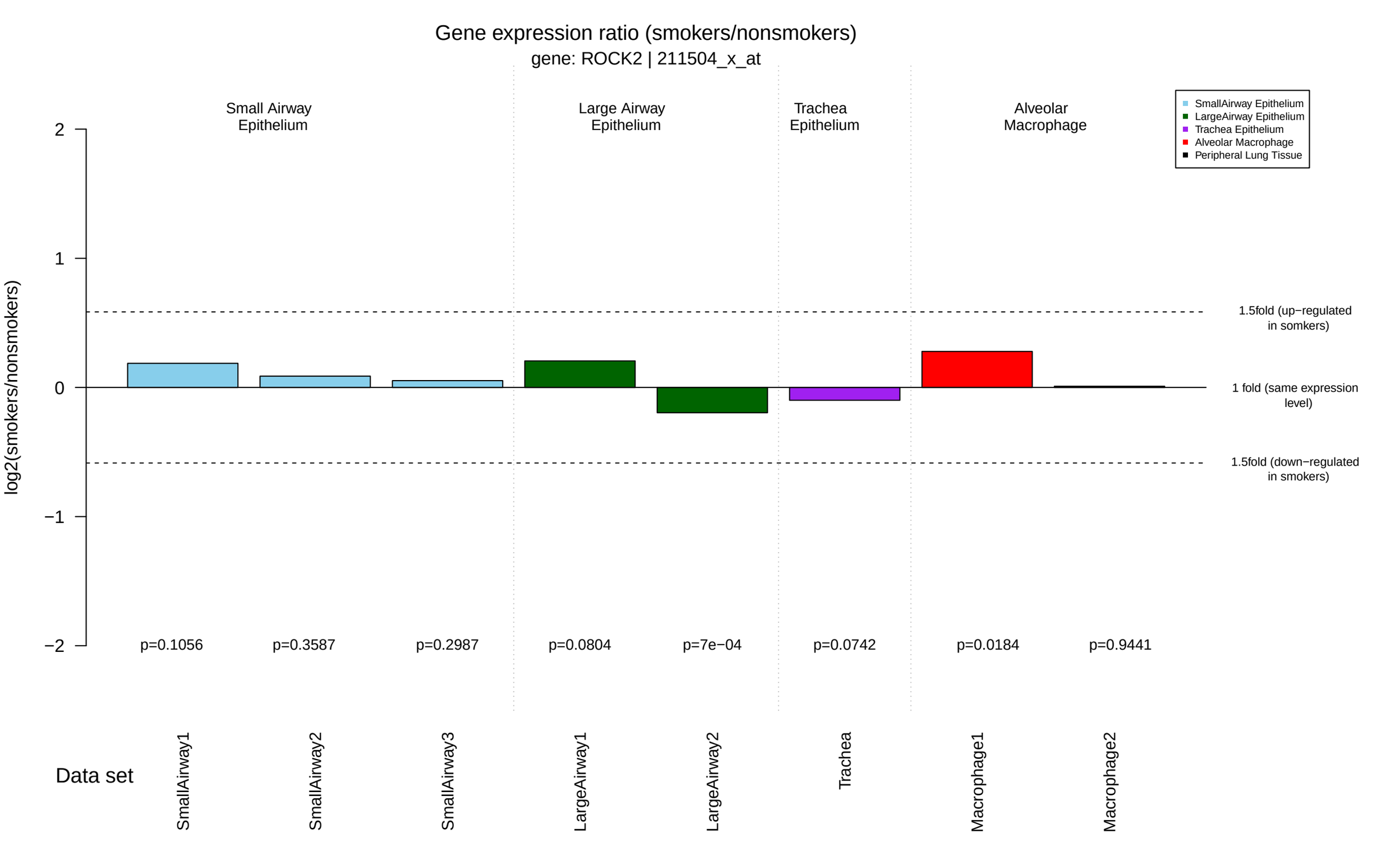
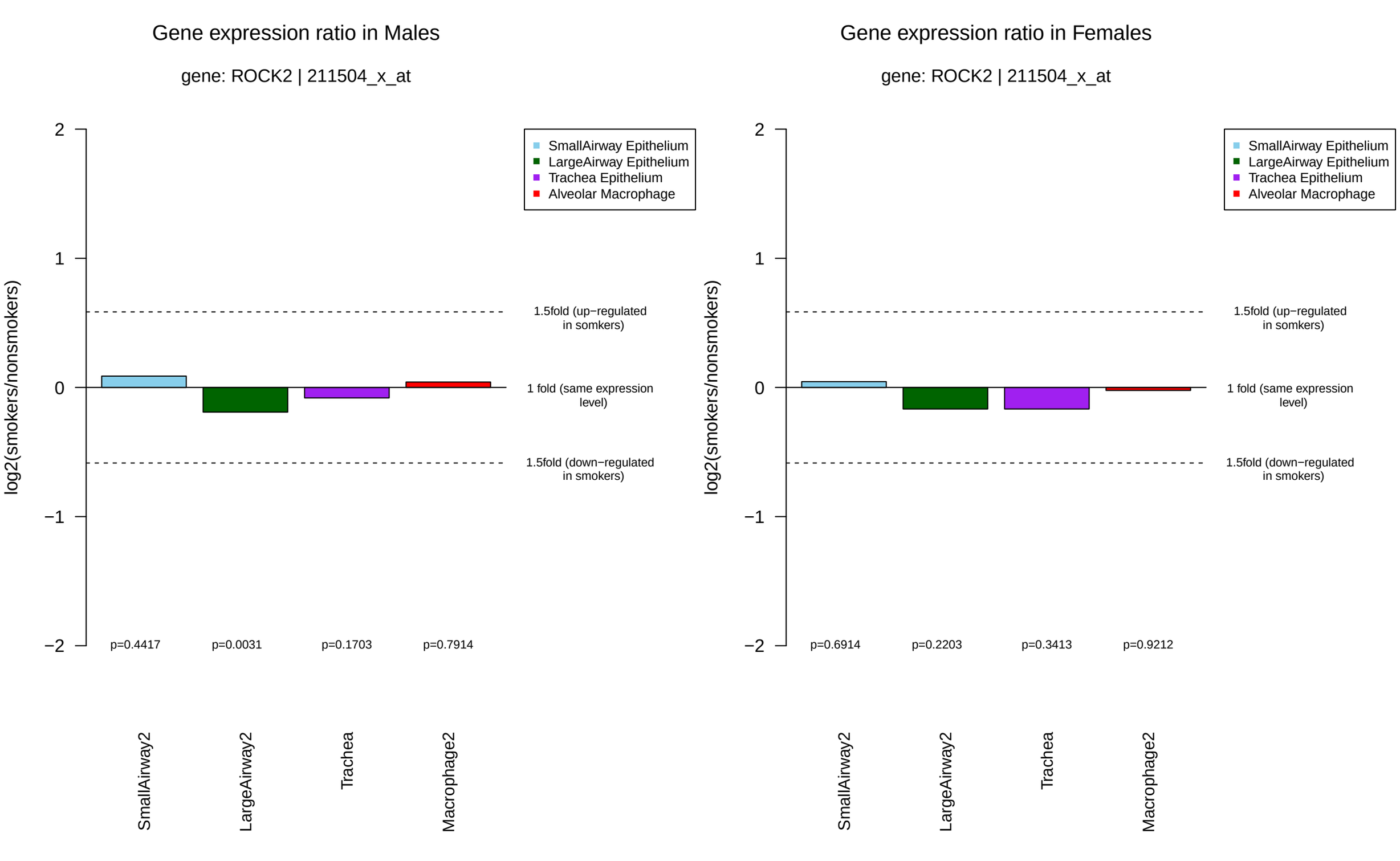
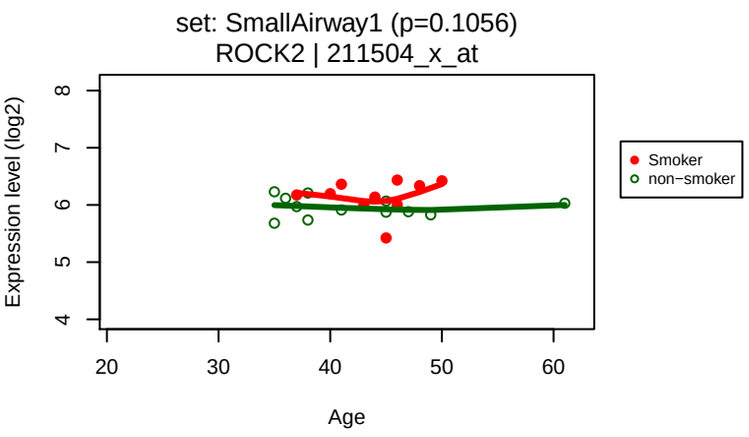
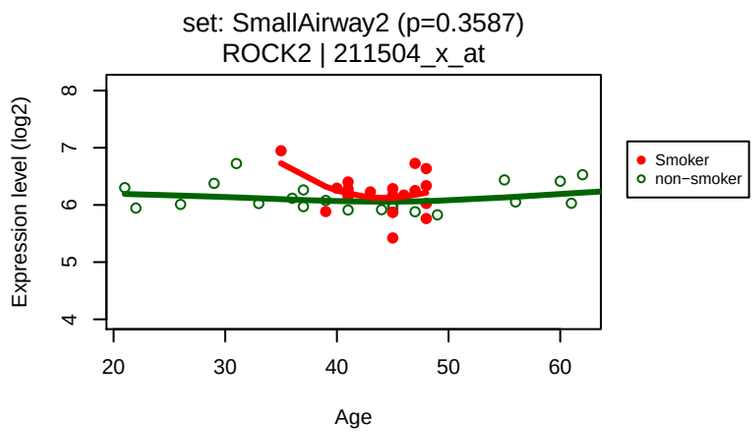
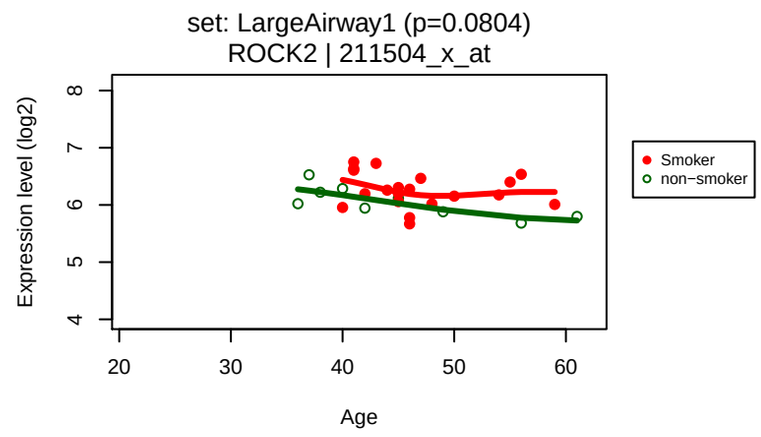

Gene expression of smokers and nonsmokers
gene: ROCK2 | 211504_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1056
p=0.3587
p=0.2987
p=0.0804
p=7e−04
p=0.0742
p=0.0184
p=0.9441
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

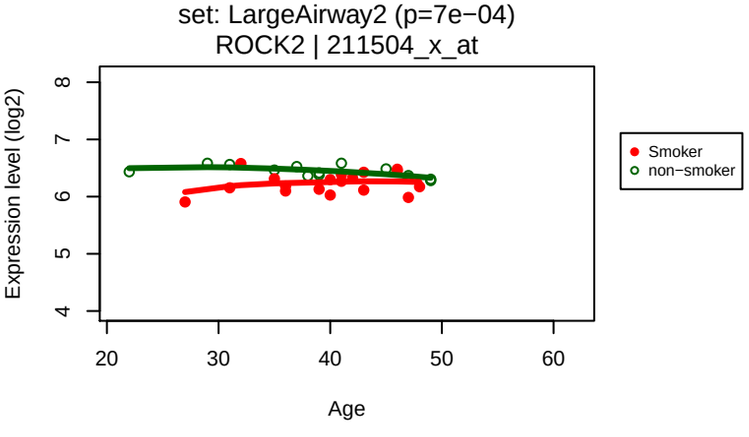

Gene expression ratio (smokers/nonsmokers)
gene: ROCK2 | 211504_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1056
p=0.3587
p=0.2987
p=0.0804
p=7e−04
p=0.0742
p=0.0184
p=0.9441
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

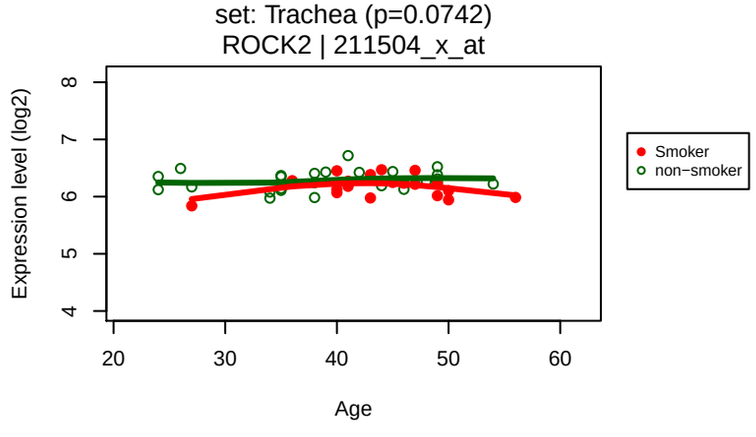

Gene expression ratio in Males
gene: ROCK2 | 211504_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4417
p=0.0031
p=0.1703
p=0.7914
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ROCK2 | 211504_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6914
p=0.2203
p=0.3413
p=0.9212
SmallAirway2
LargeAirway2
Trachea
Macrophage2

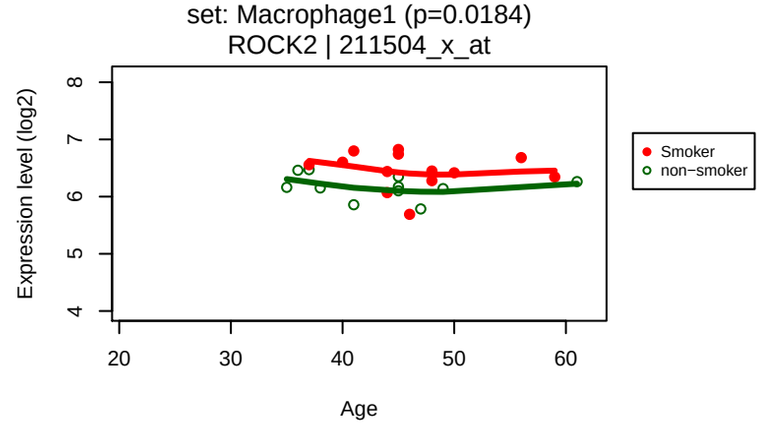

set: SmallAirway1 (p=0.1056)
ROCK2 | 211504_x_at
Expression level (log2)
Age
Smoker
non−smoker

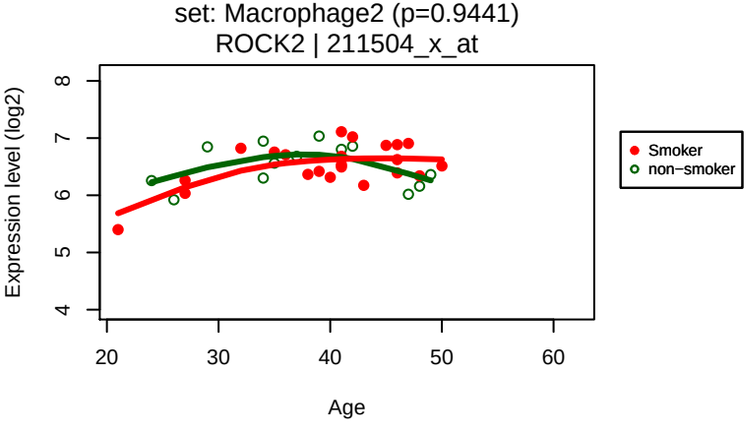

set: SmallAirway2 (p=0.3587)
ROCK2 | 211504_x_at
Expression level (log2)
Age
Smoker
non−smoker

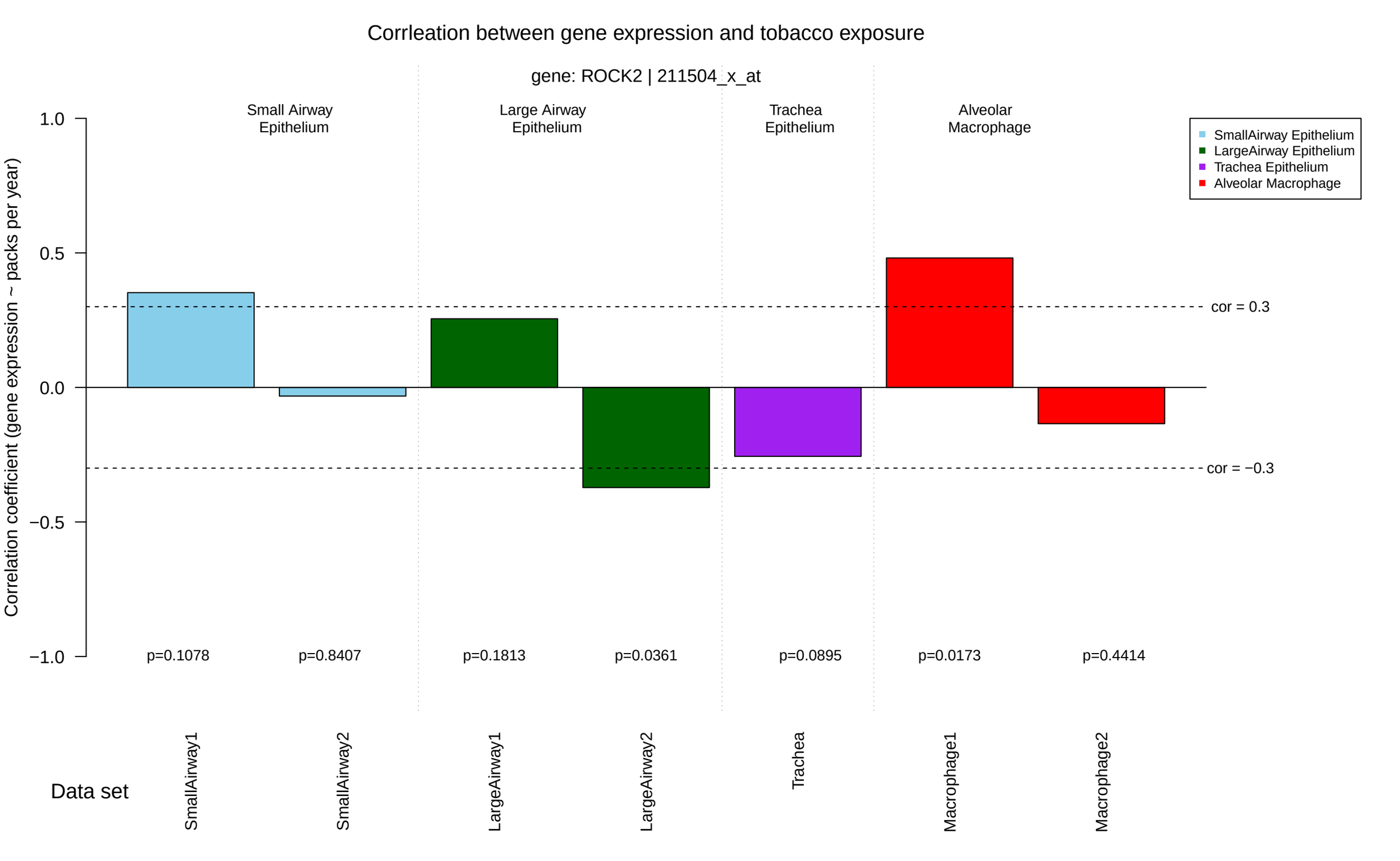

set: LargeAirway1 (p=0.0804)
ROCK2 | 211504_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=7e−04)
ROCK2 | 211504_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0742)
ROCK2 | 211504_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0184)
ROCK2 | 211504_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9441)
ROCK2 | 211504_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ROCK2 | 211504_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1078
p=0.8407
p=0.1813
p=0.0361
p=0.0895
p=0.0173
p=0.4414
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2